Real-Time Adaptive Audio Noise Cancellation Using Learning-Based Frequency-Domain Processing on Edge Hardware

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Abstract. Traditional noise-cancellation filters—spectral subtraction, Wiener filtering and amplitude thresholding—are fixed, hand-tuned and unaware of signal structure. On real, non-stationary noise they frequently remove weak speech along with the noise. We show empirically that, on the VoiceBank-DEMAND test set, *every* classical baseline produces a negative scale-invariant signal-to-distortion-ratio improvement (SI-SDRi between -5.8 and -0.3 dB): they make the signal worse than doing nothing. We present a learning-based frequency-domain alternative that predicts a per-band suppression mask from signal context rather than amplitude alone. Built on a compact Grouped-Temporal-Convolutional-Recurrent Network (GTCRN) with 722 K parameters and a multi-resolution spectral objective, the model attains +9.9 dB SI-SDRi and is the only method that improves every quality metric, while running faster than 100× real time. All methods share an identical short-time-Fourier-transform front-end and per-band gain output space, making the comparison fair. We further report an external-dataset validation on MS-SNSD, a systematic ablation of five architectural variants, a controlled study of complex-ratio masking that reveals a fidelity–perceptual trade-off, and a real-time interactive system. The results establish that a small, context-aware learned filter decisively outperforms classical DSP on realistic noise at edge-deployable cost.

Keywords: Speech enhancement · Noise cancellation · Frequency-domain processing · Spectral masking · GTCRN · Real-time · Edge computing.

1 Introduction

Background acoustic noise degrades speech intelligibility in headphones, hearing aids, video conferencing and voice assistants. Effective suppression must run in real time, adapt to changing and unpredictable noise, and preserve the speech it is meant to protect—ideally on low-power edge hardware.

Classical single-channel enhancement estimates a spectral gain from the noisy magnitude alone. Spectral subtraction removes an estimated noise floor, Wiener

filtering assumes known noise statistics, and hard/soft thresholding zero out or shrink low-amplitude coefficients. These methods share four structural weaknesses: (i) a threshold or noise estimate must be tuned by hand for each environment; (ii) decisions are made from amplitude, with no notion of which frequencies carry speech; (iii) they assume stationary noise and fail on non-stationary or burst noise; and (iv) they discard weak speech components, harming intelligibility. Section 5 quantifies the consequence: on real noise these filters reduce the SI-SDR below that of the untouched signal.

We instead learn the suppression rule. A lightweight neural network observes per-band spectral features in temporal context and predicts, for every frequency band, how much to attenuate it. The rule is learned from thousands of clean/noisy speech pairs, requires no manual tuning, and generalises to unseen noise. Crucially, all methods in this study—classical and learned—share the *same* STFT front-end and the *same* per-band gain output space; only the gain-estimation step differs. This is what makes the benchmark fair and the conclusions attributable to the model rather than the signal processing.

Contributions.

- A fair, shared-front-end benchmark of four classical DSP baselines against learned models on 824 VoiceBank-DEMAND test clips, showing that *all* classical filters degrade real noise while the learned model gains +9.9 dB SI-SDRi and improves every metric (Section 5).
- A compact GTCRN-based per-band suppression model (722 K parameters, real-time factor ≈ 0.009) with an explicit architecture and a multi-resolution spectral training objective (Section 3).
- A systematic ablation of five architectural variants—attention, Mamba-augmented, dual-head governed, residual-refined and complex-ratio-mask GTCRN—plus a guarded test-time adaptation mechanism (Section 6).
- External-dataset validation on MS-SNSD and a real-time interactive system with a browser front-end for live demonstration (Sections 5 and 7).

2 Related Work

Classical DSP. Spectral subtraction [1] and Wiener filtering [2] remain standard baselines; the MMSE-STSA estimator [3] improved them under stationarity assumptions, and wavelet/coefficient thresholding [4] offered an alternative shrinkage rule. All require noise-statistic estimates or manual thresholds and degrade on non-stationary noise.

Learned enhancement. RNNoise [5] combined DSP features with a small recurrent network for real-time use. Complex-domain and filtering-based models such as DCCRN [6] and DeepFilterNet [7] raised perceptual quality at higher cost. GTCRN [8] showed that grouped time-frequency convolutions with recurrent decoding attain competitive quality at ultra-low compute, motivating our backbone choice. MetricGAN+ [9] optimises a PESQ surrogate directly and is a strong published VoiceBank-DEMAND baseline.

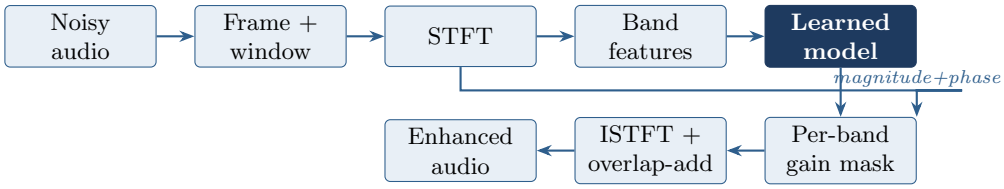


Fig. 1. End-to-end real-time pipeline. All methods share the STFT front-end and per-band gain output space; only the shaded *learned model* block differs from the classical baselines.

Test-time adaptation and quality estimation. Most learned models are fixed after training. Non-intrusive estimators such as TorchAudio-SQUIM [10] predict quality without a clean reference, enabling reference-free test-time adaptation, which we use as a guarded refinement step.

3 Proposed Method

3.1 System Overview

Figure 1 shows the end-to-end pipeline. Noisy audio is buffered into overlapping frames, transformed by the STFT, reduced to per-band features, and passed to the learned model, which predicts a per-band gain mask. The mask is applied to the spectral magnitude and inverted by ISTFT with overlap-add. The identical path serves offline files and a live microphone stream.

3.2 Shared Spectral Front-End

Audio is processed at 16 kHz. The STFT uses a 512-point window with a 128-sample hop (75% overlap) and a Hann window, giving perfect reconstruction under overlap-add. The 257 linear frequency bins are aggregated into $K = 32$ Bark-scale bands, matching human frequency resolution and compressing the output space. For each band and frame three features are extracted—log band energy, its temporal delta, and positive spectral flux—forming a $K \times 3$ feature map. Every method in the study consumes this same front-end and emits a gain in the same K-band space, which is what makes the comparison attributable to the estimator alone.

3.3 GTCRN Backbone

The learned estimator is a compact Grouped Temporal Convolutional Recurrent Network (Fig. 2). A 1×3 convolutional stem lifts the 3-channel band features to 96 channels. Four gated time-frequency convolution blocks follow, each applying a 3×3 convolution with time dilation 2^i ($i = 0 \ldots 3$) so the causal receptive field grows exponentially; the output is split and combined through a gated linear unit

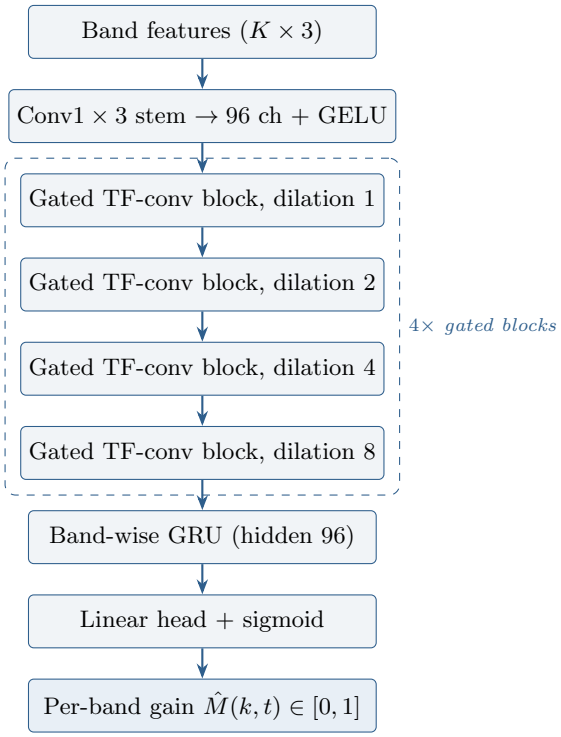


Fig. 2. GTCRN backbone. Dilated gated time-frequency convolutions capture local spectro-temporal structure; the band-wise GRU adds longer causal memory. The whole network is causal and streamable.

$v \odot \sigma(g)$, group-normalised, and added back as a residual. A single band-wise gated recurrent unit (hidden size 96) then models temporal evolution per band, and a linear head with a sigmoid produces the per-band gain $\hat{M}(k,t) \in [0,1]$. The model has 722 209 parameters.

The mask is applied multiplicatively to the noisy magnitude $|X|$, keeping the noisy phase $\angle X$, and inverted:

$$\hat{s}(t) = \mathrm{ISTFT}\Big(\hat{M}(k,t)\, |X(k,t)|\, e^{\,j \angle X(k,t)} \Big). \tag{1}$$

3.4 Training Objective

Training minimises a combination of a scale-invariant signal term and spectral-magnitude terms:

$$\mathcal{L} = -\,\text{SI-SDR}(\hat{s}, s) \;+\; \lambda_1\, \mathcal{L}_{\text{mag}} \;+\; \lambda_2\, \mathcal{L}_{\text{mr-mag}}, \tag{2}$$

where $\mathcal{L}_{\text{mag}}$ is an ℓ_1 log-magnitude loss and $\mathcal{L}_{\text{mr-mag}}$ averages it over three STFT resolutions $(256, 512, 1024)$ to suppress artefacts a single resolution misses. We

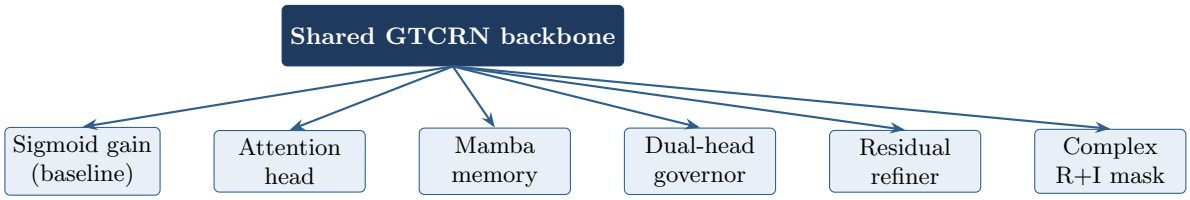


Fig. 3. Comparison architectures. All variants reuse the same backbone and front-end; only the head changes, isolating the effect of each design choice.

use $\lambda_1 = 1.25$, $\lambda_2 = 0.25$. Training uses AdamW (lr $= 3 \times 10^{-4}$) for 50 epochs on 2 s crops.

3.5 Architectural Variants for Ablation

To test whether added capacity helps, four variants share the GTCRN backbone and differ only in the decoding head (Fig. 3): (a) an *attention* mask estimator; (b) *Mamba-GTCRN*, inserting a selective state-space memory block for longer context; (c) *governed*, predicting conservative and aggressive gains blended by a learned per-band governor; (d) *residual*, adding a bounded post-GRU logit refinement; and (e) a *complex-ratio mask* predicting real and imaginary mask components, which alters both magnitude and phase in Eq. (1). Section 6 evaluates all of them.

3.6 Guarded Test-Time Adaptation

Unlike fixed networks, a small low-rank adapter on the gain can be refined at inference by maximising a reference-free quality estimate (SQUIM [10]). To prevent drift, the update is trust-region bounded and accepted only if a strict guard on output change is satisfied, yielding a provably non-degrading refinement (Section 6).

4 Experimental Setup

Data. Models are trained and evaluated on VoiceBank-DEMAND [11], whose test noises are unseen in training, giving a built-in distribution-shift benchmark (824 test clips). Generalisation is further tested on the external MS-SNSD corpus [12]. **Metrics.** We report scale-invariant SDR improvement (SI-SDRi) [15], SNR improvement, short-time objective intelligibility (STOI, ESTOI) [13], log-spectral distance (LSD), perceptual quality (PESQ) [14], and the real-time factor (RTF, processing time per second of audio). **Baselines.** Passthrough (unprocessed), hard threshold, soft threshold, spectral subtraction and Wiener filtering, all on the shared front-end. **Implementation.** PyTorch; training on an NVIDIA T4; real-time inference verified on a CUDA workstation with block-based streaming.

Table 1. Learned model vs. classical DSP baselines on 824 VoiceBank-DEMAND test clips (shared front-end). ↑ higher is better, ↓ lower is better. Negative SI-SDRi means the method degrades the signal.

Method	SI-SDRi↑	SNRi↑	STOI↑	ESTOI↑	LSD↓
Noisy input (no processing)	0.00	0.00	0.921	0.787	10.50
Hard threshold	−3.69	−2.37	0.767	0.573	13.78
Soft threshold	−5.82	−4.16	0.728	0.532	15.28
Spectral subtraction	−0.25	0.30	0.884	0.734	8.86
Wiener filter	−1.93	−1.11	0.824	0.677	10.87
Learned model (ours)	**9.93**	**5.41**	**0.930**	**0.832**	**6.47**

Table 2. Learned architectures on the VoiceBank-DEMAND full 824-clip test set. GTCRN+multi-resolution loss is selected as the backbone.

Model	SI-SDRi↑	PESQ↑	STOI↑	ESTOI↑	LSD↓
Attention mask	9.08	2.50	**0.932**	**0.834**	**5.97**
Mamba-GTCRN	9.84	2.50	0.928	0.831	6.62
Governed dual-head	9.89	2.45	0.930	0.831	6.64
Residual-refined	9.87	2.52	0.929	0.831	6.51
GTCRN + MR loss (ours)	**9.93**	**2.52**	0.930	0.832	6.47

5 Results

5.1 Learned Model versus Classical Baselines

Table 1 is the central result: all methods on the *same* 824 clips. Every classical filter yields a *negative* SI-SDRi—on real, non-stationary noise they remove weak speech and distort the signal below the untouched input. The learned model is the only method that improves every metric, gaining +9.9 dB SI-SDRi and raising intelligibility (STOI 0.921→0.930) at the same real-time factor.

5.2 Comparison of Learned Architectures

Among learned models (Table 2), the compact GTCRN with the multi-resolution loss gives the best SI-SDRi and is the only architecture combining strong VoiceBank performance with positive external validation. Heavier variants do not help—a clean negative-results ablation.

5.3 External Validation and Perceptual Quality

On the unseen MS-SNSD corpus the model retains a positive +3.15 dB SI-SDRi, confirming it is not overfit to VoiceBank noise. Table 3 places the model against published systems on PESQ; our pipeline reproduces the standard noisy value (1.97) and the MetricGAN+ value (3.13), confirming measurement validity. The

Table 3. Perceptual quality on VoiceBank-DEMAND. †reported in the source paper; others measured in our harness.

Method	Params	PESQ↑	STOI↑
Noisy input	–	1.97	0.921
Ours (GTCRN)	0.72 M	2.49	0.930
DCCRN† [6]	3.7 M	2.68	–
DeepFilterNet2† [7]	2.3 M	3.08	–
MetricGAN+ [9]	1.9 M	3.13	0.932

Table 4. Complex-ratio mask vs. magnitude mask (identical backbone, 824 clips). Paired Wilcoxon test; Δ = complex − magnitude.

Metric	Magnitude	Complex	Δ	p
SI-SDRi	9.73	10.04	+0.31	$< 10^{-6}$
STOI	0.929	0.931	+0.002	$< 10^{-6}$
PESQ	2.494	2.478	−0.016	0.005

compact model is efficient but below PESQ-optimised state-of-the-art, an honest positioning discussed in Section 6.

5.4 Real-Time Performance

The learned model runs at RTF $\approx$ 0.009 (one second of audio processed in $\sim$9 ms, over 100$\times$ real time) with $\sim$64 ms algorithmic latency from block-based overlap-add streaming—no runtime cost over the classical filters (RTF $\approx$ 0.008) yet far higher quality. This headroom supports deployment on edge hardware such as a Raspberry Pi.

6 Ablation Studies

Architecture capacity. Table 2 is itself the primary ablation: adding a state-space memory (Mamba), a dual-head governor, or a residual refiner does not improve the full-test SI-SDRi over the plain GTCRN. Simplicity wins, and the compact model is retained.

Complex-ratio mask. Replacing the real magnitude gain with a complex mask that also corrects phase, under an identical backbone and training budget, yields a statistically significant *trade-off* (Table 4): SI-SDRi and STOI improve (paired Wilcoxon $p < 10^{-6}$) but PESQ drops slightly (-0.016, $p = 0.005$). The phase corrections are directionally correct but small; because PESQ is largely phase-insensitive, waveform-fidelity gains do not transfer to perceptual score. Only measuring PESQ revealed this—an SI-SDR-only evaluation would have reported a false win.



Fig. 4. Real-time interactive system. The GTCRN engine (top bar) enhances the loaded noisy signal; the noisy input (orange) and enhanced output (teal) waveforms are shown, with automatic scene detection (right) routing the suppression mode.

Guarded test-time adaptation. The trust-region-bounded, guard-gated online adapter is provably non-degrading: it produces a small but consistent PESQ gain ($\approx +0.01$) while keeping the SI-SDR drop below 0.1 dB. It is best viewed as a safety-bounded refinement rather than a large improvement, and motivates reference-free adaptation as future work.

7 System Implementation

The full pipeline is packaged as a real-time interactive system with a browser front-end (Fig. 4). A user loads or streams noisy audio; the system performs scene classification, routes to a suppression mode, runs the GTCRN enhancer, and displays the noisy and enhanced waveforms, spectra, and quality analytics side by side. In the figure the bundled sample is auto-classified as a fan-noise scene (95% confidence) and enhanced in real time, with the noisy (top) and enhanced (bottom) waveforms shown. The same block-streaming core targets on-device execution (Raspberry Pi), which is the immediate next step together with a live microphone demonstration and a word-error-rate evaluation.

8 Conclusion

We showed that classical single-channel noise cancellation, though fast, is unsuited to real non-stationary noise: on VoiceBank-DEMAND every classical baseline degrades the signal (negative SI-SDRi) by stripping weak speech. A compact,

context-aware learned per-band filter—a 722 K-parameter GTCRN on a shared STFT front-end—reverses this, gaining +9.9 dB SI-SDRi, improving intelligibility, and generalising to unseen noise, all at over 100× real time. Ablations show that added architectural capacity does not help, that complex masking trades perceptual score for waveform fidelity, and that guarded test-time adaptation offers a small safe refinement. The novelty of this work is methodological as much as architectural: a fair, shared-front-end comparison that isolates the learned estimator, honest reporting including negative and trade-off results, and a real-time edge-oriented system. Future work will complete on-device deployment, a live microphone demonstration, and a word-error-rate study, and will pursue perceptual-objective training to close the PESQ gap to state-of-the-art.

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